

On Combining Diverse Models for Lyrics-Based Music Genre Classification

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Abstract—Automatic music organization and retrieval is a highly required task nowadays. Labeling songs with summarized but descriptive information have implications in a wide range of tasks in this scenario. The genre is one of the most common labels used for music recordings. Using this piece of information, music platforms can organize collections by, for instance, associating songs and artists with similar characteristics. Lyrics represent an alternative source of data for genre recognition. While “traditional” bag-of-words-based text mining techniques represent a considerable part of the literature, recent papers shown an advance on this task applying deep learning algorithms. However, there is no research on how these distinct strategies contribute to each other. In this paper, we explore different strategies for music genre classification from lyrics and show that even simple combinations of these strategies allow improving accuracy on the lyrics-based music genre identification.

1. Introduction

Automatic music organization and retrieval is a highly required task nowadays, in a scenario where digital music platforms are part of the day-by-day of a multitude of people. Labeling songs with summarized but descriptive information have implications in these and several other related tasks, such as automatic playlists generation, creating user profiles, and improving recommendation systems [1], [2], [3]. Genre is one of the most common labels used for music recordings. Using this piece of information, music platforms can organize collections by, for instance, associating songs and artists with similar characteristics.

However, classifying music by genre is a hard task even for human beings [4]. In this context, researchers have proposed applying an ensemble method as an alternative to overcome the subjectivity of genre information and, as consequence, improve the classification accuracy [5], [6].

One example is [6], where the authors use audio, image and text data to classify albums by genre. Specifically, they combine the classification of each track (audio), with album cover (image) and reviews (text). As this and other work claim, the accuracy in each modality of data is individually important, as well as it directly impacts the combined results.

The lyrics of the assessed songs is an alternative source of data in this scenario and may be used as a single source

of data or as combined with other types of data. Each lyric is intrinsic to its song, and enable us to ensemble the results with different classifiers methods. So, it may be directly applied to classify each recording separately, alone or combined with the audio [7], [8]. Also, differently of album reviews and other sources of data, it is available since the song is composed.

Recent literature has shown a significant advance in music classification using machine learning techniques based on deep neural networks (DNN) [9], [10]. However, this fact has not been commonly explored on lyrics data. With a few exceptions [11], most work on lyrics classification is based on “classical” techniques, such as bag-of-words-based text mining. However, to the best of our knowledge, there is no research on how these distinct strategies contribute to each other.

In this paper, we explore different strategies for music genre classification from lyrics. Moreover, we show that these strategies provide diversity in the classification scores, allowing an accuracy improvement when combined. In summary, the main contributions of this paper are:

- We experiment deep and “shallow” learning for classifying music genre by lyrics in different scenarios;
- We evaluate the combination of diverse classification strategies in both traditional ensemble methods and embedding outputs of different strategies in a unique model;
- We provide insights on how to use (or not use) ensemble learning for lyrics-based music classification;
- We demonstrate that classification model combination may improve the results of individual classifiers in the approached task.

The remaining of this paper is organized as follow: Section 2 introduces some related work on the literature. As this paper brings a wide experimental evaluation, Section 3 presents our experimental setup, including the bag-of-words-based techniques and neural architectures we applied in our experiments, as well as combination methods. Section 4 presents the results and a brief discussion on that. Finally, we conclude our work in Section 5, followed by the bibliographical references.

2. Related Work

Music classification is a widely researched topic, having a large number of papers in the literature. However, most of them do not approach genre classification based on lyrics. When some work tries to perform that task, it is usually made with bag-of-words-based [12] or other natural language processing techniques [13]. To the best of our knowledge, Tsaptsinos [11] is the only author who attempted to classify music genre with deep learning algorithms solely using lyrics. Specifically, the author used a Hierarchical Attention Network (HAN) [14], proposed for text categorization. This algorithm considers the text structure, taking into account the hierarchy term/sentence/document.

When comparing HAN to other techniques, the author report the classification models learned with this neural network achieved better results than simpler like logistic regression. However, the accuracy obtained by HAN was only slightly better than a simpler neural network, namely the Long Short-Term Memory (LSTM) [15].

Besides that, it is possible to find papers which focus on lyrics-based music classification using deep neural networks, but focusing on targets other than genre. In these cases, a common classification objective is the mood or the emotion and, usually, it is fused with the classification on the respective audio data [8], [16], [17].

3. Experimental Evaluation

This section presents an experimental evaluation of the proposed methods. For the sake of reproducibility and supplementary elements, all code and detailed results are provided in a supplementary website¹.

This section is split in two sections. First, we describe the datasets we used and how we preprocessed them for the training and classification phases. Next, we describe the experimental setup regarding the learning algorithms and how we combined results obtained by different methods. Finally, we present and discuss the results of our experiments.

3.1. Dataset

The datasets used in our experiment are diverse regarding the volume of data and class balancing. For ensuring this diversity, we use four datasets, all of them comprising seven genres. Specifically, each lyric has one label among the following values: country, folk, indie, jazz, metal, pop, and rock.

We use some preprocessing techniques to avoid noises in the lyrics. First, we set all words to lowercase. Then, we use lemmatization on the generated tokens. For neural networks, we use these tokens to create a numeric sequence as input. The dataset is not grouped by genres to avoid pitfalls during the learning process. However, we notice that, for most datasets, the classes are massively unbalanced. After the preprocessing step, we set up a cross-validation model, and

the dataset is to split into the training set, validation set, and test set.

The description of each dataset used in this research is explained next.

- **D1:** This is the largest dataset we used, composed of 201210 songs and seven distinct genres, most composed of Rock and Pop songs that implies to an unbalanced set. Specifically, these two genres comprise around 53% and 20% songs in this dataset, respectively.
- **D2:** This dataset is composed of 77735 songs by seven genres. Once again, most of songs are labeled as Pop and Rock.
- **D3:** This dataset is also composed of songs from seven genres, also it is a unbalanced dataset. The size of this dataset is 82626 songs.
- **D4:** The smallest dataset for training the model, composed of 50654 songs and seven genres, most songs labeled as Rock or Pop.
- **D5:** This dataset is a small subset of D1's songs, but the number of songs for each genre is fixed to 2000 examples enabling a balanced learning. There are only 14000 songs in total in this dataset.

3.2. Experimental Setup

In this section, we describe how we have designed our experimental setup. Specifically, we describe the algorithms used for the bag-of-words-based classification, the neural architectures applied, and the combination strategies used.

3.2.1. Traditional Text Mining Approaches. One common technique in text classification is count words frequency for each text and then build a matrix through a bag-of-words (BoW) model. This approach does not consider word order or even semantic meaning but shows words multiplicity. There are others parameters which can fit better than just frequency. The inverse of words frequency, the frequency of n-grams, a sequence of words which appears always in the same order, or even characters frequency seems to lead reasonable results.

After this step, we used the preprocessed dataset as input for five distinct learning algorithms. Specifically, we used Naive Bayes, Support Vector Machines, Linear Regression, XGBoosting and Random Forests. We only present the results obtained by Random Forest in this paper, as it performed consistently better than the other algorithms. However, we emphasize that we present detailed results in the paper's website.

3.2.2. Deep Neural Networks. Besides the structured data obtained by the process described on Section 3.2.1, we evaluated algorithms able to deal with the text as unstructured data. Specifically, we evaluated three types of techniques on deep learning: one-dimensional Convolutional Neural Networks (CNN), Long-Short Term Memory (LSTM) and attention mechanisms.

1. <http://www.dc.ufscar.br/~diego/BRACIS2019>.

We applied both CNN and LSTM techniques separately and combined in different ways. Regarding attention, we applied these mechanisms along with LSTMs. Also, we always apply an embedding layer as the first layer, where its weights came from the matrix built with word-embedding vectors, as described on Section 3.1.

For clarity, we now present all the deep neural architectures we used in our evaluation.

- **LSTM & CNN:** First attempted model use a LSTM layer, which works on understand the logical sequence of the input based on memory mechanism. Then apply three CNN and maxpooling layers, where the convolutional layer use a filter to find unique characteristics and maxpooling layer summarize and reduce the dimension of its input.

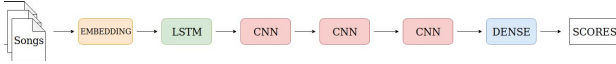


Figure 1: Architecture using a sequential LSTM and CNN layers.

- **CNN & LSTM:** In contrast, this model use CNN filters directly on the input and then apply memory mechanisms.

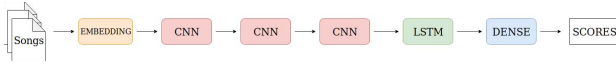


Figure 2: Architecture applying three CNN and three max-pooling layers and a LSTM after filters.

- **Merge LSTM & CNN:** This architecture use both LSTM and CNN layers separately and then merge horizontally its dense layers, applying more two dense layers before score layer.

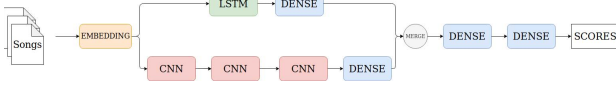


Figure 3: Architecture using a LSTM and a CNN separately, and merge technique.

- **Merge LSTM & CNN scores:** Other Architecture using merge technique but now merging scores of LSTM and CNN layers, it learn based on output scores. Also apply two dense layers before final score layer.

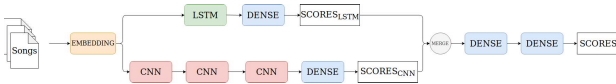


Figure 4: Architecture using other training method with merge technique.

- **Hierarchical attention:** This model use a attention layer to identify the most important words and

phrases for each class. First, it applies a bidirectional LSTM layer and the attention mechanism in all phrases separately to find words importance for each phrase. Second, repeat the layer configuration but now over the results for each phrase, trying to find phrase importance. We acknowledge the original proposal of HAN uses bidirectional Gated Recurrent Units (bi-GRU) [18], however we have evaluated with both variation and applying LSTMs performed slightly better in this case.

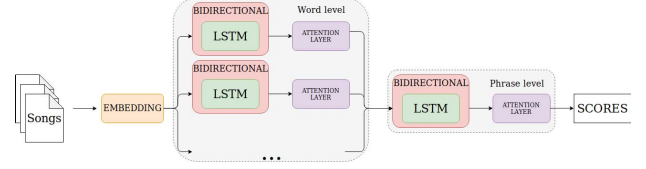


Figure 5: Hierarchical attention network.

To avoid overfitting, we used the early stopping regularization. We note that the training of all the DNN only needed a few epochs for training. Also, we trained the networks on GPUs. Even in this case, we have to acknowledge that the time for training the bag-of-words-based models is significantly lower than training these neural models.

3.2.3. Classifiers Ensembles. The main assumption of this work is that music genre classification by lyrics is a hard task to accomplish with a single classification approach, so combining the decisions of diverse classifiers improves the obtained results. To ensure our assumption is correct, we experimented on simple ensemble techniques. As show by Figure 6, we applied a parallel architecture for the combinations. In this scheme, each classifier h_i , $i \in \{1, 2, \dots, L\}$ being L the number of combined classifiers, estimates a set of scores for each class. Then, a single combiner is used to fuse these scores in a single decision.

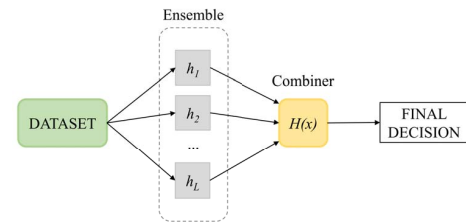


Figure 6: The parallel combination architecture, where each classifier takes an independent decision on data, and a single combiner on these decisions are applied to estimate a single label. Adapted from [19].

While we present the classifiers used in this work in Sections 3.2.1 and 3.2.2, we define the ensemble functions next.

- **Majority voting (MV):** this function performs in two steps. First, it takes a single label for each

classifier, given by $\text{argmax}(\text{scores}(h_L))$. Then, the combiner counts the voting for each class and takes the final decision by $\text{argmax}(\text{count}(C_m))$, where C_m represents each of the m possible class labels.

- **Weighted voting (WV):** while the MV considers all the classifiers with equal relevance, we also attempted to weight the votes of each model. When using WV, the importance given to its vote has a direct relationship with the estimated probability of making a correct decision. Consider p_i is the i^{th} classifier’s accuracy on a validation set. The vote of this classifier to the label it attributes to the example is then weighted (multiplied) by $w_i = p_i$. We observe that we also applied the weight defined by $w_i = \log(p_i/(1 - p_i))$, but this transformation did not present significant improvement.
- **Scores averaging (SA):** The SA strategy also combines classifications with different weights. However, the weight attribution is given song by song, according to the sets of scores obtained by each classifier. More specifically, the SA initially calculates the average score among all the classifiers for each class and then decides for the label with the highest average score value.
- **Combination learning (CL):** The CL uses a meta-algorithm for learning the combination function. In this work, we applied a stacked generalization with a single-layer fully-connected network. This strategy works using a new dataset composed of the scores for each class estimated by each base classifier. The training set for learning the combiner is obtained by reapplying the training data to the learned model and using the scores as attribute values and expect the actual genre as output. Once a new example is provided, we estimate the scores for each class and each classifier and create a new example using the same format than in the training case. Finally, the class label estimated for this new example is used as the class for the input lyrics. When applying the CL as ensemble function, it is necessary defining a learning algorithm to create a model to perform the final classification. In this work, we used the Random Forest, as it is simple and effective.

4. Results and Discussion

At this point, we are ready to present our empirical evaluation. We start by presenting the results on individual classifiers. The results on combining the classifiers are presented in the following section.

4.1. Individual Accuracy

In this section, we present the individual results of the base classifiers. Table 1 presents the accuracy obtained by each model.

In this first experiment, we can easily notice some trends in the results. For example, the accuracy values were

TABLE 1: Accuracy obtained by each individual classifier

Algorithm	Dataset				
	D1	D2	D3	D4	D5
BoW	0.646	0.608	0.604	0.523	0.434
CNN	0.582	0.526	0.529	0.422	0.416
LSTM	0.609	0.563	0.586	0.238	0.350
LSTM & CNN	0.593	0.534	0.537	0.512	0.348
CNN & LSTM	0.540	0.479	0.458	0.484	0.136
Merge LSTM&CNN	0.610	0.554	0.534	0.513	0.364
Merge LSTM&CNN 2	0.623	0.582	0.569	0.502	0.330
HAN	0.589	0.568	0.554	0.465	0.413
Best individual accuracy	0.646	0.608	0.604	0.523	0.434

higher for those sets with higher volumes of data. This fact reinforces the need for large volumes of data to obtain a good classification result, even for the classifier based on word frequency.

Perhaps the most surprising result was the fact that the Random Forest classifier obtained the best of all the results in all the data sets evaluated. As deep learning algorithms have been demonstrating excellent results on text classification, it is natural to expect a good performance of these methods. However, we take this as a positive result, since they are more interpretable and time efficient. Moreover, this paper seeks combining models for better accuracy and, in this case, it is interesting having diverse classifiers. As bag-of-words-based models have a completely different foundation if compared to deep models, having a good accuracy with Random Forest makes us believe it is a good choice to ensemble with the neural classifiers.

Regarding the neural networks, it is clear that using CNN alone or in a sequential way with LSTM causes a negative impact in the classification. That is easy to understand since CNN does not have a memory mechanism. However, it is interesting to notice that when the output of a CNN is merged with LSTM, it tends to achieve better results than the individual networks. This is even more evident in cases where LSTM presents poor results, as in the D4 data set. While LSTM must have been affected by some contrivance in the data, the act of combining it with CNN caused the accuracy to remain much closer to the best obtained in this experiment than the poor result obtained by LSTM. This is a practical motivation for combining classifiers, aiming a better accuracy in our task.

4.2. Ensemble Methods

In this section, we present the results of several combinations of classifiers. As we do not use report all possible combinations for obvious reasons, we begin this section by presenting the reasoning for choosing some combinations.

Since the bag-of-words-based algorithm presented the best results, several of the combinations use the model learned with this technique. However, for purposes of comparing its contribution to ensembles, there are similar combinations, but removing this technique. For clarity, we refer to the Random Forest on the bag-of-words representation solely as “BoW.”

Regarding the DNNs, in the individual experiments, we have already used different ways of merging the simpler types of units (CNN and LSTM) in a unique architecture. Here, we also evaluate combining them with the presented ensemble functions. For this, we consider these two algorithms separately or merged. In the second case, we opted by not using the “Merge LSTM&CNN 2” once it is similar to combine two models with stacked generalization (combination learning). Also, we notice that merging these models achieved slightly better results than applying them in sequence (LSTM-CNN and CNN-LSTM). So, we do not consider the later architecture.

Moreover, we evaluated the ensemble of all the models used in this step. In other words, we also report the accuracy obtained by combining BoW, CNN, LSTM, Merge LSTM&CNN, and HAN.

For the sake of better presentation, we split the results for each dataset. Table 2 presents the classification results on the dataset D1. The results presented in bold highlight the combinations that performed better than the best individual classifier.

TABLE 2: Accuracy of different combined algorithms on the dataset D1

Combined Algorithms	Ensemble Function			
	MV	WV	SA	CL
CNN + LSTM	0.583	0.609	0.634	0.605
CNN + LSTM + HAN	0.637	0.642	0.650	0.578
Merge LSTM&CNN + HAN	0.559	0.610	0.638	0.553
BoW + HAN	0.601	0.646	0.639	0.619
BoW + CNN + LSTM	0.610	0.649	0.624	0.674
BoW + Merge LSTM&CNN	0.573	0.646	0.606	0.631
BoW + CNN + LSTM + HAN	0.623	0.658	0.634	0.618
BoW + Merge LSTM&CNN + HAN	0.614	0.667	0.629	0.614
All classifiers	0.668	0.669	0.673	0.616

Before analyzing the outcomes, we present the results on the other datasets. So, we may investigate if there is a pattern on the results for different characteristics of data. The results on datasets D2, D3, D4, and D5 are presented in Tables 3, 4, 5, and 6, respectively.

TABLE 3: Accuracy of different combined algorithms on the dataset D2

Combined Algorithms	Ensemble Function			
	MV	WV	SA	CL
CNN + LSTM	0.504	0.563	0.580	0.551
CNN + LSTM + HAN	0.586	0.597	0.610	0.573
Merge LSTM&CNN + HAN	0.533	0.568	0.596	0.571
BoW + HAN	0.588	0.608	0.616	0.540
BoW + CNN + LSTM	0.648	0.626	0.650	0.544
BoW + Merge LSTM&CNN	0.626	0.608	0.647	0.509
BoW + CNN + LSTM + HAN	0.659	0.634	0.666	0.566
BoW + Merge LSTM&CNN + HAN	0.655	0.624	0.664	0.555
All classifiers	0.625	0.636	0.635	0.569

From these results, we can observe that the combination of only two classifiers tends not to provide improvements in accuracy when compared to the best of all individual performances. This is specially true if we apply MV as

TABLE 4: Accuracy of different combined algorithms on the dataset D3

Combined Algorithms	Ensemble Function			
	MV	WV	SA	CL
CNN + LSTM	0.570	0.586	0.578	0.584
CNN + LSTM + HAN	0.597	0.604	0.611	0.586
Merge LSTM&CNN + HAN	0.508	0.554	0.589	0.531
BoW + HAN	0.571	0.604	0.610	0.615
BoW + CNN + LSTM	0.594	0.598	0.595	0.615
BoW + Merge LSTM&CNN	0.556	0.604	0.601	0.608
BoW + CNN + LSTM + HAN	0.624	0.609	0.621	0.608
BoW + Merge LSTM&CNN + HAN	0.620	0.620	0.623	0.612
All classifiers	0.629	0.624	0.633	0.611

TABLE 5: Accuracy of different combined algorithms on the dataset D4

Combined Algorithms	Ensemble Function			
	MV	WV	SA	CL
CNN + LSTM	0.264	0.422	0.240	0.477
CNN + LSTM + HAN	0.410	0.472	0.405	0.490
Merge LSTM&CNN + HAN	0.466	0.513	0.542	0.455
BoW + HAN	0.487	0.523	0.543	0.426
BoW + CNN + LSTM	0.428	0.517	0.381	0.466
BoW + Merge LSTM&CNN	0.522	0.523	0.537	0.299
BoW + CNN + LSTM + HAN	0.494	0.532	0.472	0.475
BoW + Merge LSTM&CNN + HAN	0.541	0.548	0.554	0.429
All classifiers	0.520	0.539	0.526	0.478

TABLE 6: Accuracy of different combined algorithms on the dataset D5

Combined Algorithms	Ensemble Function			
	MV	WV	SA	CL
CNN + LSTM	0.395	0.416	0.419	0.386
CNN + LSTM + HAN	0.429	0.429	0.438	0.414
Merge LSTM&CNN + HAN	0.394	0.413	0.438	0.401
BoW + HAN	0.441	0.434	0.432	0.297
BoW + CNN + LSTM	0.436	0.437	0.436	0.329
BoW + Merge LSTM&CNN	0.403	0.434	0.426	0.268
BoW + CNN + LSTM + HAN	0.451	0.460	0.452	0.294
BoW + Merge LSTM&CNN + HAN	0.445	0.447	0.454	0.279
All classifiers	0.455	0.458	0.458	0.306

ensemble function. On the other hand, the results of the combinations of network-based models were shown to be somewhat more stable than the networks used separately, usually achieving better accuracy. However, we may note that combining the results of the simpler networks (LSTM and CNN) tends to present worst results in comparison to architectures which attempt to use them together in a single network.

When we compare the ensemble functions, we noticed that CL performed unsuccessfully. However, this does not mean that this technique should be completely discarded since there is a possibility that other learning algorithms achieve better results than Random Forest, used in our experiments. One way to understand this fact is by observing that the architecture Merge LSTM & CNN 2, which achieved better results than most individual models, is a stacked generalization based on a fully-connected network.

Besides, we notice that WV and SA outperformed the individual outcomes more frequently than MV. Also, the best result for all datasets, as well as the highest average scores, were obtained by the SA. In this way, we understand that SA, even though it is a quite simple technique, has proved to be an appropriate option for ensemble learning for the lyrics-based classification.

The combination of all classifiers, except when CL was applied as ensemble function, proved to be quite effective, obtaining the best results among all the experiments. However, we recognize that assuming this as a premise can compromise the time efficiency of the task. Therefore, we believe there is a need for algorithms able to learn better combinations and to provide small subsets of models with a potential to provide good accuracy without major losses in time performance.

5. Concluding Remarks

In this paper, we performed a wide empirical evaluation of lyrics-based music genre classification algorithms. For this, we used deep learning techniques and algorithms based on word frequency (bag-of-words). The results showed that, although simple, bag-of-words-based algorithms may surpass the results obtained by deep learning techniques.

In addition, we evaluated different techniques of combining classifiers, showing these two categories of machine learning algorithms can complement each other, achieving better results when used together. Specifically, the best results on all the evaluated datasets were obtained by combining bag-of-words with neural network-based models using a simple scoring average ensemble function.

As future work, we intend to take two different directions. One of them is the search for meta-learning algorithms capable of learning a small subset of classifiers to form a committee that provides accurate classifications without harming the time performance. Also, we intend to use the knowledge obtained in this work to create multimodal music classifiers which consider the lyrics as part of its input.

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